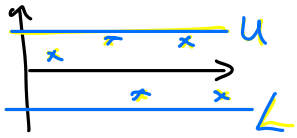
Recap: Chapter 2 - Sequences (a_n)

Bounded Sequence



Convergent Sequence



$(\forall \varepsilon > 0)(\exists N \in \mathbb{N})$
$(n > N \Rightarrow |a_n - a| < \varepsilon)$

Theorem 4.9: Uniqueness of Limits

The limit of a convergent sequence is unique.

Proof: Let (a_n) be a convergent sequence and assume towards a contradiction that $\lim(a_n) = a$ and $\lim(a_n) = \tilde{a}$ with $a \neq \tilde{a}$. Then for every $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon$$

for all $n > N_1$. Similarly, there exists $N_2 \in \mathbb{N}$ such that

$$|a_n - \tilde{a}| < \varepsilon$$

for all $n > N_2$. Let $N = \max(N_1, N_2)$, then both $|a_n - \tilde{a}| < \varepsilon$ and $|a_n - a| < \varepsilon$ for all $n > N$.

Then, $|\tilde{a}-a| = |\tilde{a}-a+a_n-a_n|$

$$= |a_n-a-(a_n-\tilde{a})|$$

$$\leq |a_n-a|+|a_n-\tilde{a}|$$

$$< \varepsilon + \varepsilon$$

$$= 2\varepsilon$$

for all $n > N$. Since this true for every $\varepsilon > 0$, let $\varepsilon = \frac{1}{3}|\tilde{a}-a| \in \mathbb{R}^+$. Substituting into the above,

$$|\tilde{a}-a| < \tfrac{2}{3}|\tilde{a}-a|$$

$$\Rightarrow \quad 1 < \tfrac{2}{3}, \text{ for all } n > N.$$

This is a contradiction. Hence, $a = \tilde{a}$. ∎

5 - Divergent Sequences

Defn 5.1 : A sequence (a_n) is said to *diverge* if it does not converge. They can be characterised into the following three forms:

i) (a_n) diverges to $+\infty$ if all $M>0$, there exists $N \in \mathbb{N}$ such that $a_n > M$ for all $n > N$.

ii) (a_n) diverges to $-\infty$ if for all $M<0$, there exists $N \in \mathbb{N}$ such that $a_n < M$ for all $n > N$.

iii) Otherwise, (a_n)'s limit does not exist.

Remark: The following notation is used for divergent sequences

i) $\lim(a_n) = \infty$, $\lim(a_n) = -\infty$.

ii) $a_n \to \infty$ or $a_n \to -\infty$.

iii) $\lim_{n\to\infty}(a_n) = \infty$, $\lim_{n\to\infty}(a_n) = -\infty$.

iv) $a_n \to \infty$ as $n \to \infty$, $a_n \to -\infty$ as $n \to \infty$.

Ex 5.3: i) See notes.

ii) Let (b_n) be the sequence given by $b_n = (-1)^n$.

Then (b_n) diverges.

Proof: Suppose towards a contradiction that (b_n) converges to $b \in \mathbb{R}$. Let $\varepsilon = 1/2$. Then, there must exist an $N \in \mathbb{N}$ such that

$$|(-1)^n - b| < 1/2$$

for all $n > N$. For each even $n > N$, it follows that

$$|1 - b| < 1/2 .$$

Similarly, for each odd $n > N$, it follows that

$$|-1 - b| = |1 + b| < 1/2 .$$

Hence,

$$\begin{aligned} 2 &= |1 + b + 1 - b| \\ &\leq |1+b| + |1-b| \\ &< 1/2 + 1/2 \\ &= 1 \end{aligned}$$

for all $n > N$. This is a contradiction and hence (b_n) is divergent. ∎

Theorem 5.4 : Convergence and Bounded

All convergent sequences (a_n) are bounded.

Proof : Let $\lim(a_n) = a$. That is, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon$$

for all $n > N$. Choosing $\varepsilon = 1$ and defining the set

$$S = \{|a_1|, |a_2|, \ldots, |a_N|, |a+1|, |a-1|\}.$$

Letting $K = \max(S)$ then it follows, by construction of S, that

$$|a_n| \leq K$$

for all $n \in \mathbb{N}$. Hence, (a_n) bounded. ∎.

Ex 5.5 : Let (a_n) be a sequence given by $a_n = (-1)^n$. Then (a_n) is divergent.

Proof : This sequence is shown to be unbounded in Example 3.3 (ii). Hence, by the above Theorem, (b_n) is

divergent. [2]